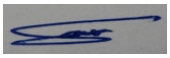
Program and Branch Name:
[B.Tech CSE-F]

PBL Progress Report – I

Field	Details
Course Name	Data Structure and Algorithms - II
Course Code	CCSE0301
Faculty Name	Mr. Shamshad Ali
Assignment Type	Individual Assignment
Student Name	Mohd Sabir Ansari
ERP ID (	0261DCS032
Team/Group No. (Group/Capstone only)	Not Applicable - Individual Assignment
Problem Statement / Project Title	Intelligent Education Resource Sharing System
SDG(s)	SDG 4 — Quality Education , SDG 17 — Partnerships
Reporting Month	Month 1
Date of Submission	16/09/2026

Student Declaration

I certify that this assignment is my own work, with all sources appropriately acknowledged. I understand that a 0% mark and further academic penalties may apply if the content is found to be plagiarised or inappropriately AI-generated.

Student Name & Signature: Mohd Sabir Ansari &

Date: 16/09/2026

PBL Progress Report – I

Review 1: Problem Understanding & Initial Progress

Review Coverage: Individual Assignment: Unit/Module 1 (Trees) and Unit/Module 2 (Graphs)

<table>
<tr><th>Field</th><th>Details</th></tr>
<tr><td>1. Problem Statement</td><td>Intelligent Education Resource Sharing System

The Intelligent Education Resource Sharing System is proposed to provide a centralized platform for efficiently organizing, searching, accessing, and sharing educational resources such as notes, lectures, assignments, and e-books. The system will use Tree and Graph data structures to organize resources and represent relationships between users, subjects, topics, and resources.</td></tr>
<tr><td>2. Problem Context & Background</td><td>Educational resources like notes, lectures, assignments, and e-books are often scattered across different platforms, making them difficult to organize and find. The Intelligent Education Resource Sharing System provides a centralized platform to upload, organize, search, share, and access study materials based on subject, topic, and resource type, with Trees used for efficient organization.</td></tr>
<tr><td>3. Problem Identification / Understanding</td><td>The initial study shows that educational resources are often scattered and difficult to access. The system will help students, teachers, and administrators upload, organize, search, and share resources. Trees can be used for structured resource organization, while Graphs can represent relationships between users, subjects, topics, and resources.</td></tr>
<tr><td>4. Objectives</td><td>• To understand the problems in traditional educational resource sharing and identify its limitations.
• To study and apply Tree- and Graph-based DSA concepts for organizing resources and representing their relationships.
• To improve the searching, sorting, retrieval, and sharing of educational content through a structured system.</td></tr>
<tr><td>5. Stakeholders</td><td>Students, teachers, faculty members, and academic administrators who use and manage educational resources.</td></tr>
<tr><td>6. Research / Literature Review Completed</td><td>1. H. Chen and Y. Shang, “Construction and Application of a Learning Resource Sharing Platform in Higher Education Based on Mobile Interactive Technology,” International Journal of Interactive Mobile Technologies (iJIM), vol. 20, no. 1, pp. 19–33, 2026, doi: 10.3991/ijim.v20i01.59785.

2. C. Meng, C. Pan, Z. Li, C. Zhou, X. Cao, and J. Zhu, “DFRP: A Dual-Track Feedback Recommendation System for Educational Resources,” in Proceedings of the Thirty-Third International Joint Conference on Artificial Intelligence (IJCAI-24), Demonstrations Track, pp. 8741–8744, 2024.</td></tr>
<tr><td>7. Unit/Module Concepts Identified for Application</td><td>
<table>
<tr><th colspan="2">DSA Concept</th><th>Possible Application</th><th>Time Complexity</th></tr>
<tr><td colspan="2">Binary Search Tree</td><td>Store resources according to resource ID, title, or topic</td><td>O(log n) average, O(n) worst case</td></tr>
<tr><td colspan="2">Tree Traversal</td><td>Display educational categories and resources systematically</td><td>O(n)</td></tr>
<tr><td colspan="2">AVL Tree</td><td>Maintain balanced resource records during frequent insertion and deletion</td><td>O(log n)</td></tr>
<tr><td colspan="2">Adjacency List / Matrix</td><td>Represent relationships among users, subjects, topics, and resources</td><td>AdjacencyMatrix: O(V²), AdjacencyList: O(V + E)</td></tr>
<tr><th>DSA Concept</th><th>Why It Is Not Used in Our System</th><th>Where It Can Be Used</th><th>Time Complexity</th></tr>
</table>
</td></tr>
</table>

	Binary Tree	BST and AVL are more suitable for our requirements.	Hierarchical data representation	O(n) traversal
	Memory Representation of Tree	Low-level memory representation is not required.	Array or linked-node implementation	O(n) space
	Pre-order Traversal	Root-first processing is not required.	Tree copying and hierarchical display	O(n)
	Post-order Traversal	Child-first processing is not required.	Tree deletion and expression evaluation	O(n)
	Constructing Binary Tree from Traversals	The system does not reconstruct trees from traversal sequences.	Reconstructing trees from traversal data	O(n)
	Binary Heap	Heap-based processing is not required.	Priority-based processing	O(log n)
	Threaded Binary Tree	Threaded tree structure is not required.	Efficient tree traversal	O(n)
	Priority Queue	Priority-based processing is not required.	Task scheduling	O(log n)
	Heap Sort	Heap-based sorting is not required.	Sorting large datasets	O(n log n)

8. Work Completed During Month 1	During Month 1, the project problem was finalized and the issues in conventional educational resource sharing were identified. The target users, stakeholders, and major resource types were defined. Unit 1 Trees and Unit 2 Graphs were studied, and their possible use in resource categorization, searching, and relationship mapping was identified. An initial conceptual design and plan for future development were also prepared.
9. Evidence	The project is supported by the Unit 1 and Unit 2 PPTs provided by the university for DSA-II, along with relevant research papers on educational resource sharing and recommendation systems. These sources helped in understanding the concepts and developing the proposed system. https://drive.google.com/drive/folders/1vC6Y4zEfRpabYIPz2I724XI-gl4uMTWK Github Repository - https://github.com/sabiransari786/Intelligent-Eductation-Resource-sharing-system
10. Challenges Faced	The main challenges were applying Tree and Graph concepts, selecting suitable data structures, and organizing resources clearly while keeping the system design simple.
11. Individual / Team Contribution	• I defined the project problem, analyzed the resource-sharing issues, and identified user requirements. • I studied Unit 1 (Trees) and Unit 2 (Graphs) and mapped these concepts to the proposed system. • I prepared the initial project structure, documentation, and plan for Review 2.
12. Overall Progress (%)	30%
13. Plan for Next Review	In the next review, the system requirements and suitable data structures will be finalized. Basic modules for resource sharing, categorization, and searching will be developed using Tree and Graph concepts. The initial features will be tested, and supporting material such as diagrams, code samples, and screenshots will be collected for Review 2.

Faculty Evaluation:

- To be completed by Faculty in Moodle